

Shijie Xiao

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RESEARCH INTEREST: Computer Vision, Generative Model, AI for Climate, Physics Informed ML

EDUCATION

Georgia Institute of Technology, Atlanta, GA Aug 2025-May 2029
PhD in CS advised by Josiah Hester and Ali Sarhadi Focus on Physics-ML hybrid model for weather forecasting

Georgia Institute of Technology, Atlanta, GA May 2026
Master of Science in Computer Engineering GPA: **3.90/4.00**

Relevant Coursework: **Computer Vision, Adv Prog Techniques**, Generate and Geometric DL, ML for Neural/Behavioral Data

Huazhong University of Science and Technology, Wuhan, China June 2024
Bachelor of Science in Artificial Intelligence & Automation GPA: **3.95/4.00**

SKILLS

Programming Python, C++&C, Java, Bash, MATLAB, Linux **AI Frameworks** PyTorch, Tensorflow, OpenCV
Language Mandarin (**Native**), English **Version Control** Git, Excel, SQL **Model** Diffusion Model, Transformer, CNN, LSTM

EXPERIENCE

Graduate Research Assistant with Prof. Ali Sarhadi sponsored by Sandia National Lab, Georgia Tech May 2025-Current
FAST-ML: A Hybrid Physics–Machine Learning Framework for Tropical Cyclone Intensity Forecasting (for JAMES)

- Develop FAST-ML Framework: embed neural networks into physics models to build a hybrid physics-ML system.
- Engineer Differentiable Optimization-Learn dynamic environmental parameters directly from 3D atmospheric data,
- Outperform Google SOTA in some cases while preserving physical interpretability.

Physics-based Generative AI for Probabilistic Forecasting of Hurricane-Driven Compound Flooding (for NMI)

- A TF-based generative emulator translates tropical cyclone tracks into compound flood fields.
- Overcame historical data scarcity by training on massive ensembles of physics-simulated synthetic tracks coupled with complex hydrodynamic and hydraulic simulations.
- Slashed computational bottlenecks by generating probabilistic flood maps from ECMWF ensembles in seconds instead of days, enabling real-time, risk-informed emergency planning.

Research Intern & Master Thesis with Prof. Anqi Wu, Georgia Tech | **GNN, Behavior Analysis** Jan-Dec 2025

A Computational Framework for Multi-Agent Behavior Analysis: Geometric Vision Extraction and Social Attention Inference

- Conducted research on machine learning methods for trajectory analysis to understand animal behavior patterns.
- Implemented algorithms to predict mice movement and modeled spatiotemporal interactions using graph structures.
- Developed training pipelines and introduced a novel geodesic loss function that improved performance.

Research Assistant & UG Thesis at HKUST with Prof. Yincong Chen | Stable Diffusion, Image Segmentation Jan-June 2024

Enhancing segmentation from segmented datasets containing noisy labels using generative models

- Conducted research on applying noisy label segmentation to diffusion models for image generation tasks.
- Proposed a novel approach for learning from noise-infused labels in both unsupervised and semi-supervised settings.
- Utilized unlabeled and noisy-labeled image datasets to improve segmentation performance using Stable Diffusion.

Research Assistant at CUHK with Prof. Wu Yuan | Diffusion Models, OCT Imaging Dec 2022-Mar 2023

Multiplicative speckle noise denoising of OCT medical images based on Diffusion Model

- Developed a DDPM-based denoising method for speckle noise in OCT medical images to improve diagnostic accuracy.
- Implemented noise modeling, diffusion training pipelines, and enhanced convergence speed and prediction efficiency.
- Reproduced baseline models including DN-CNN, U-Net, BM3D, TCFL-OCT, and DDPM for comparative evaluation.
- Wrote Python scripts to compute denoising metrics such as PSNR, SSIM, and GCMSE.

Participant at CVPR 2023 BDD100K Challenges | Object Tracking, Autonomous Driving, DETR, YOLO-NAS May-June 2023

Multi Object track to guide autonomous driving through machine learning algorithms

- Implemented object detection with advanced SOTA detectors including YOLO and DINO.
- Developed a multi-object tracking pipeline for autonomous driving using state-of-the-art machine learning models.
- Ranked in the top 20% for Multi-Object Tracking (MOT) and achieved first place in Object Detection (OD) tasks.

Undergrad Research Assistant at HUST with Prof. Xiang Bai | OCR, MMOCR, DBNet Sep-Dec 2022

Recognition and extraction of musical notes using Optical Character Recognition (OCR)

- Conducted research on text detection of musical notation using Optical Character Recognition (OCR) techniques.
- Built and annotated a specialized dataset from online musical tune data for model training and evaluation.
- Evaluated models including DBNet, DBNet++, and PANet using the MMOCR framework and detection libraries.

PROJECTS

eSports Gaze Behavior Detection | CNN, DETR, Few-Shot Learning Feb 2022-April 2023

- Designed and developed a personalized eSports assistant using deep learning for gaze tracking across video frames.
- Built a cross-frame gaze estimation model using convolutional neural networks and temporal feature alignment.
- Proposed a novel few-shot learning framework allowing for rapid adaptation to users with minimal calibration data.
- Final system supports real-time inference and operates with <50ms latency on standard GPUs.

Health Management System with Health Code and Trip Card | SQL, C++, Data Structure June 2020-May 2021

- Developed a health code system, including QR code generation, data management, and epidemic risk level updates.
- Designed a structured user database supporting registration, trip tracking, and automated status updates.
- Implemented modular system features such as location-based risk evaluation and dynamic color coding.